

Ejercicio 1

```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 10.16.35.2 255.255.255.192 10.16.35.1
Checking for duplicate address...
PC1 : 10.16.35.2 255.255.255.192 gateway 10.16.35.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.16.35.2/26
GATEWAY    : 10.16.35.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC1>

```

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Executing the startup file

PC2> ip 10.16.35.66 255.255.255.224 10.16.35.65
Checking for duplicate address...
PC1 : 10.16.35.66 255.255.255.224 gateway 10.16.35.65

PC2> save
Saving startup configuration to startup.vpc
done

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.16.35.66/27
GATEWAY    : 10.16.35.65
DNS        :
MAC        : 00:50:79:66:68:01
.PORT      : 10034
RHOST:PORT : 127.0.0.1:10035
RTU:       : 1500

PC2> █

```

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Press '?' to get help.

Executing the startup file

PC3> ip 10.16.35.98 255.255.255.240 10.16.35.97
Checking for duplicate address...
PC1 : 10.16.35.98 255.255.255.240 gateway 10.16.35.97

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> show ip

NAME       : PC3[1]
IP/MASK     : 10.16.35.98/28
GATEWAY     : 10.16.35.97
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 10030
RHOST:PORT  : 127.0.0.1:10031
MTU         : 1500

PC3>

```

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Press '?' to get help.

Executing the startup file

PC4> ip 10.16.35.114 255.255.255.248 10.16.35.113
Checking for duplicate address...
PC1 : 10.16.35.114 255.255.255.248 gateway 10.16.35.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 10.16.35.114/29
GATEWAY    : 10.16.35.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC4>

```

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```

R1
PC1 PC2 PC3 PC4 R1 R2 R3
R1(config)#
*Mar 1 00:11:28.995: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:11:29.995: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config)#end
R1#wr
*Mar 1 00:11:46.395: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface 2/0
^
% Invalid input detected at '^' marker.

R1(config)#interface FastEthernet 2/0
R1(config-if)#ip address 10.16.35.121 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:13:13.835: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Mar 1 00:13:14.835: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R1(config)#end
R1#write
*Mar 1 00:13:23.031: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.16.35.1 YES manual up up
FastEthernet0/1 10.16.35.65 YES manual up up
FastEthernet1/0 10.16.35.97 YES manual up up
FastEthernet2/0 10.16.35.121 YES manual up up
R1#
```

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```

PC1 PC2 PC3 PC4 R1 R2 R3
Mar 1 00:00:05.547: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
R2#enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#no ip domain-lookup
R2(config)#hostname R2
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 10.16.35.122 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface
Mar 1 00:11:24.243: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
Mar 1 00:11:25.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#interface FastEthernet0/1
R2(config-if)#ip address 10.16.35.125 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
Mar 1 00:13:04.519: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
Mar 1 00:13:05.519: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R2(config)#end
R2#write
Mar 1 00:13:12.679: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#show ip address brief
^
% Invalid input detected at '^' marker.

R2#show ip interface brief
interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.16.35.122    YES manual up          up
FastEthernet0/1 10.16.35.125    YES manual up          up
R2#
```

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```

PC1 PC2 PC3 PC4 R1 R2 R3
*Mar 1 00:00:04.415: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Mar 1 00:00:04.419: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Mar 1 00:00:05.399: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet
et0/0, changed state to down
*Mar 1 00:00:05.399: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet
et0/1, changed state to down
R3#enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#no ip domain-lookup
R3(config)#hostname R3
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.16.35.126 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:12:31.507: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:12:32.507: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#interface FastEthernet0/1
R3(config-if)#ip address 10.16.35.113 255.255.255.248
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar 1 00:14:02.975: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:14:03.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R3(config)#end
R3#wri
*Mar 1 00:14:06.331: %SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
R3#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/0    10.16.35.126    YES manual up          up
FastEthernet0/1    10.16.35.113    YES manual up          up
R3#
```

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```
PC1 PC2 PC3 PC4 R1 R2 R3
PC1> ip 10.16.35.2 255.255.255.192 10.16.35.1
Checking for duplicate address...
PC1 : 10.16.35.2 255.255.255.192 gateway 10.16.35.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.16.35.2/26
GATEWAY    : 10.16.35.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC1>
PC1>
PC1>
PC1> ping 10.16.35.66
10.16.35.66 icmp_seq=1 timeout
84 bytes from 10.16.35.66 icmp_seq=2 ttl=63 time=31.966 ms
84 bytes from 10.16.35.66 icmp_seq=3 ttl=63 time=31.609 ms
84 bytes from 10.16.35.66 icmp_seq=4 ttl=63 time=31.261 ms
84 bytes from 10.16.35.66 icmp_seq=5 ttl=63 time=31.607 ms

PC1> trace 10.16.35.66
trace to 10.16.35.66, 8 hops max, press Ctrl+C to stop
 1  10.16.35.1  15.857 ms  15.046 ms  15.459 ms
 2  *10.16.35.66  30.619 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> 
```

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① 10.16.39.0 /24

00/05/24/24
16 59

① nova mascara

★) $2^6 - 2 \geq 50$

$32 - 6 = 126$

255.255.255.192

Δ) $2^7 - 2 \geq 6$

$32 - 3 = 129$

255.255.255.248

□) $2^5 - 2 \geq 25$

$32 - 5 = 127$

255.255.255.224

X, ◇) $2^2 - 2 \geq 2$

$32 - 2 = 130$

255.255.255.252

○) $2^4 - 2 \geq 12$

$32 - 4 = 128$

255.255.255.240

saltos

64

32

16

8

4

4

red	dir	PHost	UHost	broadcast
★	10.16.39.0	10.16.39.1	.62	.63
□	10.16.39.64	10.16.39.65	.94	.95
○	10.16.39.96	10.16.39.97	.110	.111
Δ	.112	10.16.39.113	.118	.119
X	.120	10.16.39.121	.122	.123
◇	.124	10.16.39.125	.126	.127
	.128			

disf	IP	gateway	Pc1
R1	10.16.39.1	-	0/0
Pc1	10.16.39.2	10.16.39.1	710
R1	10.16.39.65	✓	210
Pc2	10.16.39.66	10.16.39.65	010
R1	10.16.39.97	-	011
Pc3	10.16.39.98	10.16.39.97	010
R3	10.16.39.113	-	011
Pc4	10.16.39.114	10.16.39.113	010
R1	10.16.39.121	-	011
R2	10.16.39.122	-	010
R2	10.16.39.125	-	011
R3	10.16.39.126	-	010
R3	10.16.39	-	011

192.16.19.0

Ejercicio 2

192, 16, 35, 0/24

000 524/24
16 35

* $2^6 - 2 \geq 50$
 $= /26$
 255, 255, 255, 192

D) $2^5 - 2 \geq 25$
 $= /27$
 255, 255, 255, 224

C) $2^4 - 2 \geq 12$
 $= /28$
 255, 255, 255, 240

Δ $2^3 - 2 \geq 6$
 $= /29$
 255, 255, 255, 248

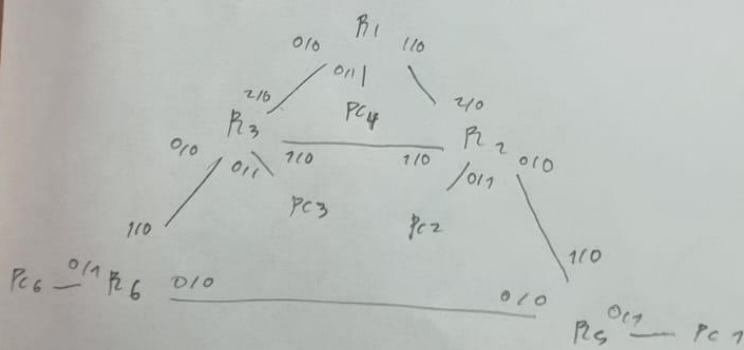
A, B, C, D, E, F) $2^2 - 2 \geq 2$
 $= /30$
 255, 255, 255, 252

Salto5

- 64
- 32
- 16
- 16
- 8
- 4
- 4
- 4
- 4
- 4

$\diamond$ $2^4 - 2 \geq 8$
 $= /28$
 255, 255, 255, 240

disP	IP
R5	192.16.35.1
PC1	.2
R2	.65
PC2	.66
R3	.97
PC3	.98
R1	.113
PC4	.144
R4	.129
PC6	.130



gateway

